ABSTRACT

A smoke detector is a device that senses smoke typically as an indicator of fire. Smoke detector basically works on principle of optical detection or on ionization and for increased sensitivity for smoke detection both the methods are used .By this smoke detectors can be used to detect the smokers in public places. However these smoke detectors in large commercial industrial and residential are usually provided by a central fire alarm system. smoke detection is done by MQ2 analog gas sensor. This sensor is also used to detect LPG, i-butane ,propane, methane ,alcohol, hydrogen ,smoke. There by when ever smoke is detected by this sensor it gives information to arduino, and the programme in the Arduino runs and gives the information that smoke detected and buzzer buzzes in caution of smoke. Coming to design of smoke detectors, these are typically housed in a disc shaped plastic enclosure about 150mm in diameters and 25mm in thick, but shape can vary by manufacturer or product line.

INTRODUCTION

The IoT Smoke Detection System combines the power of an Arduino Uno microcontroller with key components including an MQ-2 gas sensor, Liquid Crystal Display (LCD) with I2C interface, and a buzzer. This integrated system is designed to detect the presenceof smoke and provide immediate alerts through both visual and audible means.

BACKGROUND

Air quality has become a growing concern worldwide due to its direct impact on human health and the environment. Poor air quality, characterized by the presence of pollutants and harmful gases, can lead to a range of health issues, including respiratory diseases, cardiovascular problems, and even premature mortality. Monitoring air quality is essential for public health and environmental protection, as well as for ensuring workplace safety in industrial settings.

Traditional air quality monitoring systems are often large, expensive, and limited to specific locations. They require professional installation and maintenance, making them inaccessible for many individuals and communities. In response to these limitations, there has been a growing interest in developing portable and affordable air quality monitoring solutions that can be used in homes, schools, workplaces, andot her indoor environments.

The advancement of microcontroller technology and the availability of low-cost gas sensors have made it possible for individuals and small-scale initiatives to create DIY air quality monitoring systems.

OBJECTIVES OF AIR QUALITY MONITORING SYSTEM

- Design and implement air quality monitoring system using internet of things technology
- Utilize gas sensors to calculate the percentage of particular gas in the surrounding air anddisplay it on the liquid crystal display.
- Help to control the presence of the various harmful gases in the surrounding air.
- Provide a cost effective solution to the human society.

APPLICATIONS OF AIR QUALITY MONITORING SYSTEM

Indoor Air Quality Monitoring:

Monitor indoor air quality in homes, offices, schools, and public buildings to ensure a healthy and comfortable environment. Detecting harmful gases like carbon monoxide (CO) and propane (C3H8) can prevent health risks.

Workplace Safety:

Enhance workplace safety by detecting hazardous gases in industrial settings, laboratories, and manufacturing facilities. This ensures a safe working environment for employees.

Gas Leak Detection:

Detect gas leaks in residential and commercial areas, including kitchens (e.g., LPG leaks),helping to prevent accidents and fires.

Environmental Monitoring:

Collect data on outdoor air quality in urban and rural areas to assess pollution levels and theirimpact on the environment and public health.

Smart Home Integration:

Integrate the air quality monitoring system with smart home automation to control ventilation, airpurification, and HVAC systems based on real-time air quality data.

Health and Wellness:

Monitor air quality in healthcare facilities, particularly in areas where patients with respiratoryconditions are treated. Maintain high air quality standards for patient well-being.

Research and Citizen Science:

Contribute to air quality research and citizen science projects by collecting and sharing data withscientists and environmental organizations.

Urban Planning:

Assist urban planners and policymakers in making informed decisions about infrastructure development and traffic management to mitigate air pollution in cities.

Vehicle Emissions:

Monitor air quality near busy roadways and transportation hubs to assess vehicle emissions andtheir impact on local air quality.

Emergency Response:

Provide real-time data to emergency responders during incidents involving hazardous materials,helping them assess risks and take appropriate action.

Air Quality Alerts:

Generate alerts or notifications to inform users when air quality levels deteriorate beyond safethresholds.

Home Safety:

Ensure the safety of residential spaces by monitoring for potential sources of indoor pollution,such as gas leaks or the presence of volatile organic compounds (VOCs)

INTRODUCTION TO IoT (Inernet of Things)

In an increasingly interconnected world, the Internet of Things (IoT) has emerged as a transformative force that is reshaping the way we live, work, and interact with our environment. IoT technology is a powerful fusion of sensors, connectivity, data analytics, and automation, enabling everyday objects and devices to become "smart" and communicate with each other and with us over the internet. This technological revolution has far-reaching implications for various industries and aspects of our lives.

Key Elements of IoT:

Sensors and Devices:
At the heart of IoT are sensors and devices equipped with various sensors, such as temperature,humidity, motion, and more. These devices collect data from the physical world.

Connectivity:
IoT devices are connected to the internet or other networks, allowing them to transmit data in real-time. Common connectivity options include Wi-Fi, cellular networks, Bluetooth, and low-power, wide-area networks (LPWANs).

Data Analytics:
The data collected by IoT devices is processed and analyzed to extract meaningful insights.Advanced analytics, including machine learning and artificial intelligence (AI), are often employed to derive valuable information from the data.

Cloud Computing:
Cloud platforms are used to store, process, and manage the vast amounts of data generated by IoT devices. Cloud services provide scalability, accessibility, and computational power for IoTapplications.

User Interfaces:
IoT applications often feature user-friendly interfaces, including mobile apps and webdashboards, that allow users to interact with and control connected devices.

Applications of IoT:
IoT technology finds applications in various domains, including:

Smart Home:
IoT-enabled devices in homes provide automation and control over lighting, thermostats,security systems, and appliances, enhancing convenience and energy efficiency.

Healthcare:
Wearable IoT devices and remote monitoring systems enable continuous health tracking, earlydisease detection, and improved patient care.

Industrial IoT (IIoT):
In industries, IoT technology optimizes operations through predictive maintenance, assettracking, and real-time monitoring of equipment and processes.

Agriculture:
IoT sensors and data analytics help farmers monitor soil conditions, weather, and crop health,leading to more efficient and sustainable farming practices.

Smart Cities:
IoT is used to create smarter urban environments by improving traffic management, wastemanagement, public safety, and energy efficiency.

Environmental Monitoring:
IoT sensors are deployed to monitor air quality, water quality, and wildlife habitats, aiding inconservation efforts and pollution control.

Challenges and Considerations:
While IoT offers immense potential, it also presents challenges related to data security, privacy, interoperability, and scalability. Effective IoT solutions must address these issues to ensure trustand reliability.

In summary, the Internet of Things is a dynamic and transformative technology that is shaping the future of connectivity and automation. It has the potential to create more efficient, sustainable, and responsive systems in various domains, making it a pivotal force in the ongoingdigital revolution.

LITERATURE REVIEW

"A review of gas sensors employed in electronic nose applications" by Marco S. Dutra (2012)

This paper introduces the concept of electronic noses, sensor arrays mimicking the human olfactory system for gas and odor detection. It surveys various gas sensors crucial for electronic noses, including metal oxide sensors, conducting polymer sensors, and surface acoustic wave sensors. The paper extensively covers the working principles, sensitivity, selectivity, advantages, and limitations of each sensor type. Applications in environmental monitoring, food quality control, industrial safety, and healthcare highlight the diverse utility of electronic noses.

"Internet of Things for environmental monitoring applications: A review" by Jorge SáSilva (2016)

Providing a comprehensive overview, this paper explores the application of the Internet of Things (IoT) in environmental monitoring. It discusses the integration of IoT technologies for real-time data collection, sharing insights into its potential in monitoring diverse environmental parameters. The paper addresses key aspects, including sensor deployment, communication protocols, and data analytics, offering valuable insights for enhancing environmental monitoring practices.

"A survey of wireless sensor network-based air pollution monitoring systems" bySudanthi Wijewickrama (2018)

This survey emphasizes the critical importance of monitoring air quality and pollution levels. Traditional methods are deemed insufficient, leading the paper to advocate for wireless sensor networks (WSNs). The authors highlight WSNs as cost-effective and scalable solutions for air pollution monitoring. The interconnected sensor nodes enable efficient data collection and transmission, addressing limitations associated with traditional approaches.

"Air quality monitoring system: A review" by J. H. Yoo (2008)

This review delves into air quality monitoring systems, addressing various aspects such as air pollutant measurement, sensor technologies, data collection methodologies, and applications. The paper emphasizes the continuous and accurate monitoring of air quality for public health andenvironmental protection. It

discusses challenges in air quality monitoring and explores potential solutions, providing a comprehensive overview of the field.

"A comprehensive review on smart sensors for air quality monitoring" by Valentina Parvulescu (2019)

Focusing on smart sensors for air quality monitoring, this comprehensive review explores recent advancements in sensor technology. The authors cover innovations in sensing materials, miniaturization, and communication protocols. The paper offers insights into the evolving landscape of sensor development and its applications, contributing to the understanding of emerging technologies in air quality assessment

WORKING

The "IoT-Based Smoke Detection System" presents an innovative solution to detect the presence of smoke in the surrounding air or environment by enhancing traditional systems by incorporating Internet of Things (IoT) principles. Smoke-related incidents pose significant threats to safety and property, necessitating a responsive and interconnected solution.The presence of smoke in surrounding air or environment impact public health and the environment. This project addresses this challenge by implementing the sensor nodes that detects the presence of smoke in real-time and transmits the data to a central server via robust communication protocols. The data is then processed, analyzed, and visualized through a liquid crystal display, allowing stakeholders to access information about the presence of smoke in the surrounding environment. The system also incorporates an alert mechanism that notifies users through buzzer. These sensor nodes are integrated with microcontrollers, which facilitate data collection and communication. When amount of smoke surpasses safe levels it provide the alerts to the stakeholders through visual or through alerting. Ultimately, the "IoT-Based Smoke Detection System" showcases the potential of IoT technology in addressing critical hazardous environmental challenges. The project's significance lies in its ability to provide real-time, accessible, and actionable information about smoke, fostering a more environmentally conscious society and contributing to the creation of a healthier, safe and sustainable future.

PROBLEM STATEMENT

The increasing urbanization and growing complexity of modern cities have led to a range of challenges, including traffic congestion, inefficient energy usage, and inadequate waste management. These issues not only affect the quality of life for urban residents but also have significant environmental and economic implications.

To address these challenges and create more livable and sustainable cities, there is a need for innovative solutions that leverage technology and data. This problem statement focuses on developing a Smart City solution

PROPOSED SOLUTION

The proposed "IoT-Based Smoke Dtetction System" represents a approach to encounter the growing crisis of increasing amount of smoke in the air harnessing the power of Internet of Things (IoT) technology.

This system aims to create a comprehensive solution that continuously monitors the amount of smoke in air in real time, processes the data, and provides valuable insights to users.

The system integrates a sensor and other IoT component. The sensor typically works on the principle of 'semiconductor gas sensing (MQ2/MQ135) to get the smoke parameters surrounding it.

The system then alerts the user about the presence of smoke surrounding it , and displays the realtime data on 16X2 LCD Display and provide alert through buzzer.

WORKING

Smoke Detection:

Utilizes an MQ-2 smoke sensor to continuously monitor the air for smoke particles.Analog values from the sensor are processed by the Arduino Uno microcontroller.

Alert Generation:

When smoke levels exceed a predefined threshold, the system triggers visual alerts on the LCD and audible alerts through the buzzer.The LCD displays informative messages about the detectedsmoke, enhancing user awareness.

User Interface and Display:

The Liquid Crystal Display (LCD) serves as the user interface, providing clear and real-time information about the smoke detection status.Informative messages and status updates are displayed on the LCD.

Buzzer Activation:

The buzzer is activated simultaneously with the LCD alert, providing an audible signal to promptimmediate attention.

Arduino Uno Controller:

Acts as the central microcontroller, coordinating sensor readings, alert generation, and display functions.Executes the programmed logic for monitoring and responding to smoke levels.

Specifications

Specifications	Description
Arduino Uno Integration	Utilize Arduino Uno as the control board for the system
Gas sensors	Detects the gases present in the surrounding air
Liquid Crystal I2C	Display the real time data of smoke present in the surrounding air
Buzzer	Alerts the use about the presence of smoke
IoT integration	Part of the Air Quality Monitoring System
Wiring Infrastructure	High quality efficient wiring for reliable connections and power distribution

PROCESS OVERVIEW

Initialization:
The system initializes upon power-up, with the Arduino Uno and connected components starting their operation.

Smoke Detection:
The MQ-2 smoke sensor continuously monitors the air for smoke particles.Analog readings fromthe smoke sensor are processed by the Arduino Uno, providing real-time data on smoke levels.

Threshold Comparison:
A predefined smoke threshold is set within the Arduino code.The system compares the real-time analog values with the predefined threshold to determine the presence of smoke.

Alert Generation:
When smoke levels surpass the threshold, the Liquid Crystal Display (LCD) is activated to display informative messages. Simultaneously, the buzzer is triggered to emit audible alerts, enhancing user awareness.

User Interface:
The LCD serves as the user interface, displaying clear messages about the detected smoke and system status. Informative messages keep users informed about the ongoing status of the smoke detection system.

Arduino Uno Control:
The Arduino Uno acts as the central controller, executing programmed logic to coordinate the smoke sensor, LCD, and buzzer. The microcontroller makes decisions based on the smoke levelsand activates the appropriate alert mechanisms.

MODEL OF THE PROJECT

PREPARED MODEL



fig. model

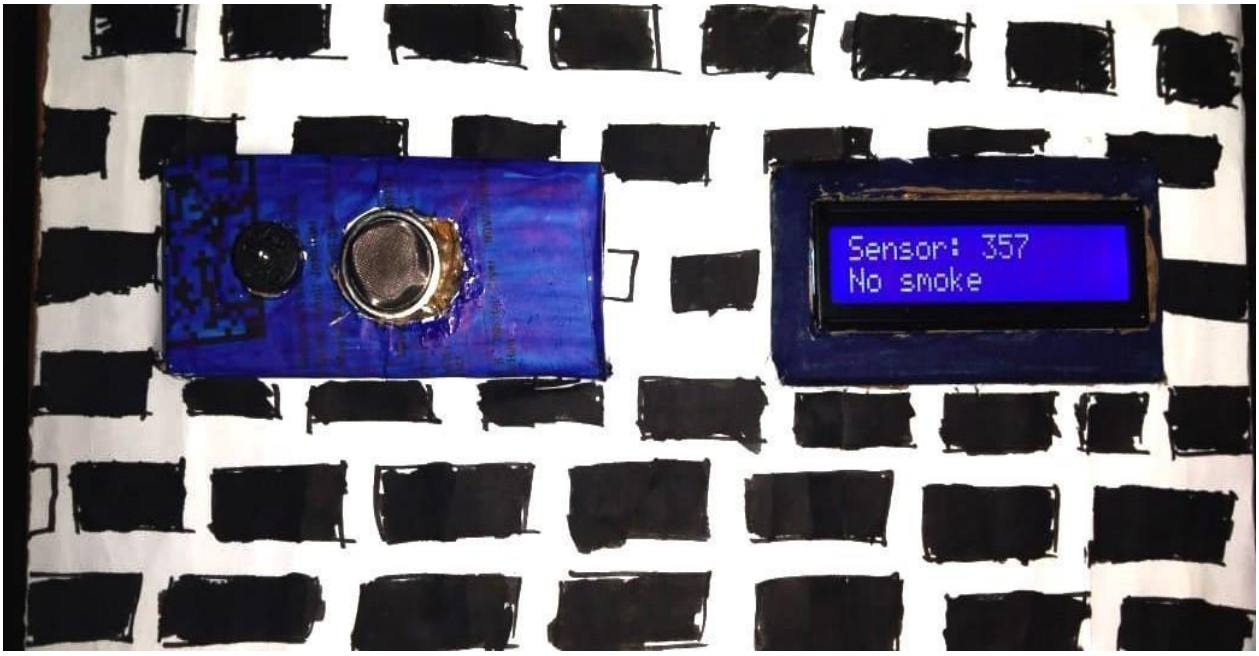


fig. model

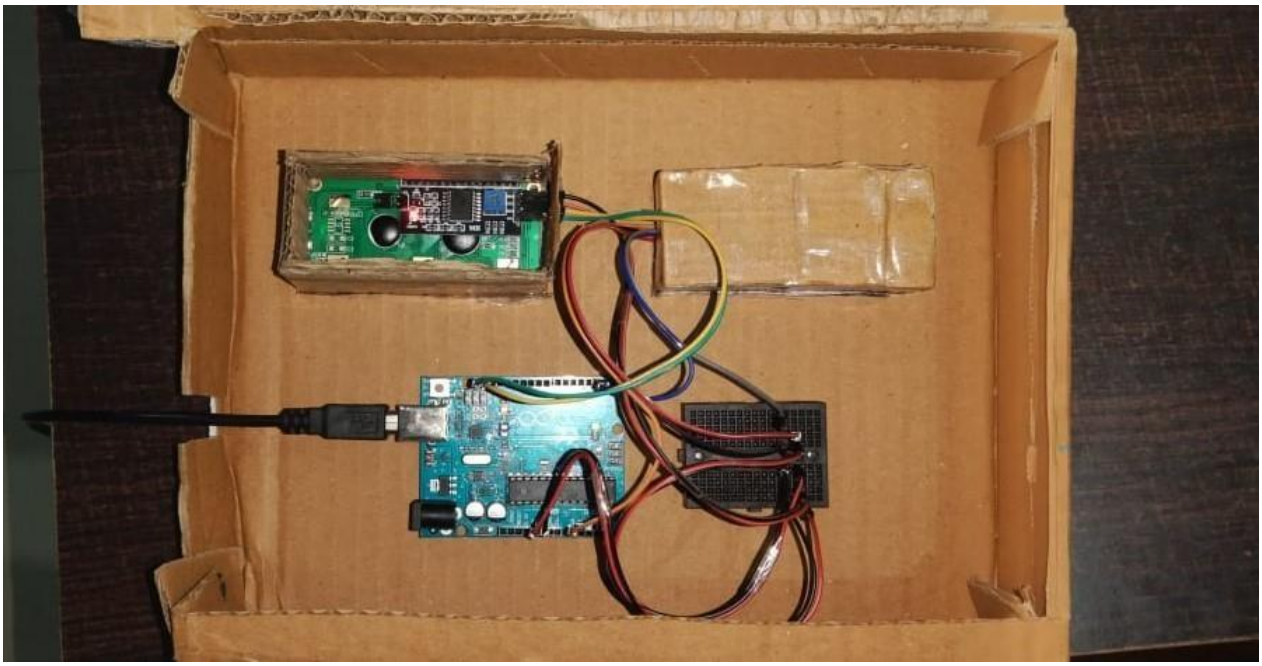


fig. model

5.1 CIRCUIT DIAGRAM

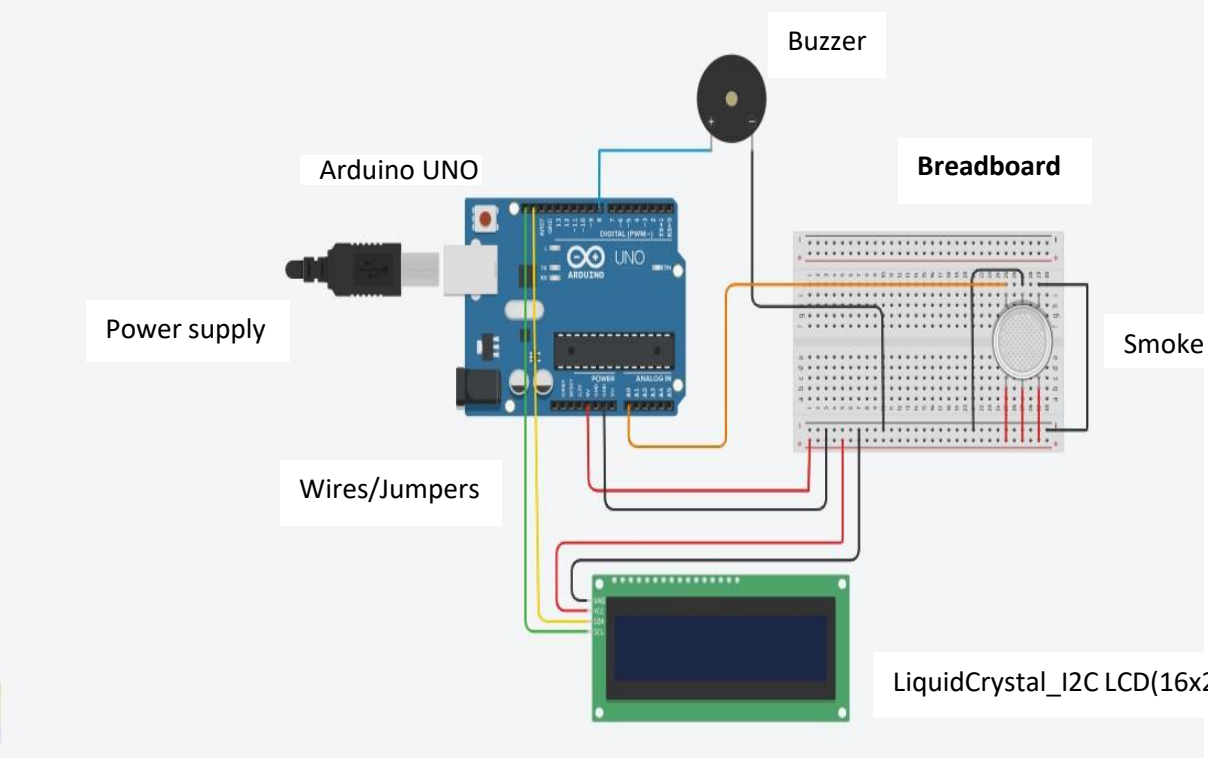


fig. circuit diagram

5.2 CODE

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2); // Replace 0x27 with your actual I2C address
int mq2Pin = A0;
//buzzerPin = 2;
int thresholdValue = 500; // Adjust this value based on sensor calibration

void setup() {
  Serial.begin(9600);
  lcd.begin(16, 2);
  pinMode(mq2Pin, INPUT);
  pinMode(2, OUTPUT);
   lcd.init();                    // initialize the lcd
  lcd.init();
  // Print a message to the LCD.
  lcd.backlight();
  lcd.setCursor(0,0);
  lcd.print("Smoke Detector");
  delay(2000);
  lcd.clear();
}
void loop() {
  int sensorValue = analogRead(mq2Pin);
  lcd.clear();
```

```
 lcd.setCursor(0, 0);
 lcd.print("Sensor: ");
 lcd.print(sensorValue);
 lcd.setCursor(0, 1);

 if (sensorValue > thresholdValue) {
  lcd.print("Smoke detected!");
   digitalWrite(2, HIGH);  // turn the LED on (HIGH is the voltage level)
   delay(1000);
 } else {
  lcd.print("No smoke");
  digitalWrite(2, LOW);  // turn the LED off by making the voltage LOW
 delay(1000);
 }
 Serial.print("Sensor Value: ");
 Serial.println(sensorValue);

 delay(1000);
}
```

MATERIALS USED AND COST ESTIMATION

ARDUINO UNO:



fig. Arduino UNO board

The Arduino Uno is a popular and versatile microcontroller board that serves as the central brain of the air quality monitoring system. Arduino Uno offers sufficient processing power to collect and process data from sensors and perform calculations required for gas concentration measurements. The Arduino Uno can be easily programmed using the Arduino IDE (Integrated Development Environment), making it accessible to both beginners and experienced developers. The code for the air quality monitoring system is uploaded to the Arduino Uno to control its operation.

The Arduino Uno's user-friendly interface and wide community support make it an ideal choice for DIY projects like air quality monitoring, allowing enthusiasts and developers to create custom electronic systems with ease. Its open-source nature fosters innovation and encourages the sharing of knowledge and code among makers and researchers.

GAS SENSORS



fig. MQ-2 gas sensor

The gas sensors used in the ‘Smoke Detection System’ are capable of detecting smoke present in the surrounding air. The MQ-2 gas sensor is designed to detect General combustible gases along with the smoke

LIQUID CRYSTAL I2C



fig. LiquidCrystal_I2C LCD(16x2)

The Liquid Crystal I2C module, often referred to as an I2C LCD module, is a crucial component for displaying information in projects. Its primary purpose is to provide a user-friendly interface by showing text and numeric data on a liquid crystal display (LCD) screen. The I2C protocol is a serial communication standard that enables devices, like the LCD module and the Arduino microcontroller in the project, to communicate seamlessly. The "I2C" in its name stands for "Inter-Integrated Circuit," emphasizing its role in interconnecting integrated circuits.

Buzzer



fig. buzzer

A buzzer or beeper is an audio signaling device, which may be mechanical, electromechanical, or piezoelectric (piezo for short). Typical uses of buzzers and beepers include alarm devices, timers, train and confirmation of user input such as a mouse click or keystroke.

BREADBOARD



fig. mini breadboard

A breadboard, also known as a prototyping board or solderless breadboard, is a fundamental tool in electronics for building and testing electronic circuits without the need for soldering orpermanent connections.

WIRES



fig. wires/jumpers

Wires play crucial role in any IoT system by connecting various components like gas sensors, liquid crystal I2C. Wires facilitate data transmission,power distribution and communication, allowing the system to collect data from sensors control lighting levels and respond to changing conditions ensuring efficient and automated quality control monitoring system.

CONCLUSION

The "IoT Smoke Detection System with LCD Display and Buzzer Alert" represents asignificant advancement in enhancing traditional smoke detection mechanisms. By integrating anArduino Uno with an MQ-2 smoke sensor, Liquid Crystal Display (LCD), and a buzzer, thesystem demonstrates a robust solution for real-time smoke monitoring and immediate user alerts.The modular design of the system allows for flexibility and scalability, accommodating potential future enhancements such as IoT integration for remote monitoring and mobile app accessibility. The combination of visual alerts on the LCD and audible alerts through the buzzerensures swift and effective communication in the presence of smoke, fostering a proactive response to potential fire hazards.

This project not only addresses the critical need for improved safety measures in various environments but also showcases the adaptability and potential for innovation in the realm of smart and interconnected security systems. The integration of modern technologies not only enhances the accuracy of smoke detection but also contributes to ease of use, accessibility, and user awareness.

In conclusion, the "IoT Smoke Detection System with LCD Display and Buzzer Alert" stands as a practical and cost-effective solution, making strides towards safer living and working environments. The system's modular architecture, coupled with the potential for future enhancements, positions it as a valuable asset in the landscape of smart safety systems, contributing to a safer and more secure future.

REFERENCES

BOOKS

5.2.1 "Internet of Things for Smart Cities: Technologies and Protocols" by Pethuru Raj, Anupama C. Raman

5.2.2 "Environmental Monitoring with Arduino: Building Simple Devices to Collect Data Aboutthe World Around Us" by Emily Gertz

5.2.3 "Wireless Sensor Networks for Environmental Monitoring" by Anna Forster, ChristopherR.H.Thompson

WEBSITE

5.2.4 IoT for all (https://www.iotforall.com/) A website dedicated to IoT applications , includingsmart city projects.

5.2.5 IEEE IoT initiative (https://iot.ieee.org) Provides resources and publications related toIoT.

5.2.6 GitHub (https://github.com) A platform where we can find open-source IoT projects andcode examples